Economic Competitiveness

of the Swiss Confederation

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Abstract. This report provides a comprehensive, multi-layered analysis of Switzerland's economic competitiveness from the 1990s to the present. Drawing on macroeconomic indicators (TFP, human capital, R&D intensity, patent density), industry-level productivity data, trade statistics from the World Bank WITS database, and empirical policy evaluations, we argue that Switzerland's sustained competitive advantage rests on four mutually reinforcing pillars: *(i)* world-leading innovation capacity and R&D investment (3.4% of GDP, 4th OECD); *(ii)* a post-industrial processing model that transforms imported raw inputs into high-value-added exports in pharmaceuticals, gold refining, and biotechnology; *(iii)* deep global trade integration anchored by EFTA, more than 120 bilateral agreements with the EU, and the 2013 Sino-Swiss Free Trade Agreement; and *(iv)* structural policies—low corporate taxation, SNB monetary stewardship, and targeted innovation subsidies—that collectively amplify firm-level investment and innovation performance. Empirical evidence from Arvanitis et al. (2010), using propensity score matching, confirms that public R&D subsidies causally raise firm innovation output. A SECO natural-experiment study (Eufinger et al., 2021) shows that the 2015 SNB rate shock boosted cumulative firm investment by 8 percentage points and employment by 7.5 percentage points. Despite strong-franc headwinds and rising Chinese competition in third-country markets, Switzerland has maintained frontier-level labour productivity ($104/hour), a near-2.9% (ILO measure) unemployment rate, and export volumes of $419.9 bn in 2023.

Contents

1. Introduction: Theoretical Framework and Swiss Context

The determinants of national and firm-level competitiveness have occupied a central position in international economics since at least Porter (1990), whose "diamond model"—factor conditions, demand conditions, related and supporting industries, and firm strategy—provided a systematic framework for understanding why certain nations excel in particular industries. More recently, the new new trade theory pioneered by Melitz (2003) complemented this approach by emphasising firm heterogeneity: only the most productive firms self-select into export markets, and exposure to international competition raises within-industry average productivity through market-share reallocation and learning-by-exporting effects. Endogenous growth theory (Romer, 1990) further underlines that economies sustaining long-run competitive advantage must invest continuously in the production of non-rival ideas—i.e., in R&D and human capital—rather than relying on factor accumulation alone.

Switzerland represents an exceptional laboratory for applying all three frameworks simultaneously. As a small, landlocked, resource-poor economy—41,285 km^2, 8.9 million inhabitants, four official languages—it has paradoxically achieved one of the highest GDP per capita levels in the world (~$99,500 in 2023, fourth globally) and maintained an average registered unemployment rate of 2.0% in 2023 (SECO definition), corresponding to an ILO-harmonised rate of approximately 2.9%, both among the lowest in the OECD (SECO, 2024; IMD, 2024). This "Swiss paradox"—extraordinary competitive success without natural resource endowments or population scale—demands careful explanation.

1.1 The Small Open Economy Constraint and its Productive Response

Small open economies (SOEs) face an inherent structural constraint: domestic market size is insufficient to exploit economies of scale across all industries. This forcess firms to internationalise early—what the international business literature terms the "born-global" imperative. For Switzerland, this constraint has historically functioned as a disciplinary spur to specialisation in high-skill, high-value-added niches where scale economies are less decisive than knowledge intensity, brand equity, and process precision. The pharmaceutical, precision instruments, and luxury watchmaking industries are perfect examples of this logic. Moreover, Switzerland's non-membership in the European Union, despite deep bilateral integration via more than 120 sectoral agreements, has endowed its monetary authority—the Swiss National Bank (SNB)—with an independent monetary policy. This independence has been both a competitive asset, protecting Switzerland from Eurozone monetary cycles, and a persistent vulnerability, exposing Swiss exporters to franc appreciation shocks (the so-called *Frankenschock* episodes of 2011 and 2015).

1.2 Four Structural Pillars of Swiss Competitiveness

Drawing on IMD (2024) and the analysis developed in the subsequent sections, we identify four interlocking structural pillars:

1. **R&D Ecosystem.** World-class research anchored by ETH Zurich and pharmaceutical

giants Novartis and Roche. Switzerland ranks fourth among OECD nations for R&D intensity at 3.4% of GDP (CHF 24.6 billion), with private enterprises self-financing 86% of their own R&D (OFS, 2023).

2. **Post-Industrial Processing Model.** Switzerland systematically imports raw and semi-finished inputs, transforms them through knowledge-intensive industrial processes, and re-exports as high-value goods—gold refining (99.99% purity), pharmaceutical formulation, and biotech manufacturing are perfect examples of the model.
3. **Deep Global Trade Integration.** Total exports reached $419.9 billion in 2023, reaching 225 countries and territories. Switzerland holds the world's highest patent application density per capita and maintains EFTA membership, bilateral EU agreements, and strategic FTAs with China and Japan (EPO, 2024; World Bank, 2023).
4. **Smart Structural Policies.** Low corporate taxation at cantonal and federal levels, targeted innovation subsidies (Innosuisse), labour market flexibility, and an open financial system attract FDI, with inward stock exceeding 105% of GDP (OECD, 2024).

Figure 1 below summarises the causal chain linking these pillars to observable export competitiveness—what we term the "Swiss Formula".

Figure 1 The "Swiss Formula": causal chain from structural policy to export competitiveness. *Authors' synthesis of OFS (2023), OECD (2024), World Bank (2023).*

The remainder of this report proceeds as follows. Section 2 analyses macroeconomic competitiveness through TFP, labour productivity, R&D, and human capital lenses. Section 3 disaggregates competitiveness to the industry level. Section 4 examines trade integration, tariff policy, and regional agreements. Section 5 evaluates structural policies and their empirical effects on firm competitiveness. Section 6 assesses China's role since 2000. Section 7 concludes.

2. Macroeconomic Competitiveness: Productivity, Innovation, and Human Capital

2.1 Total Factor Productivity: Historical Trajectory and Frontier Dynamics

Total Factor Productivity (TFP) measures output growth not attributable to factor accumulation and is therefore the key indicator of an economy's capacity to generate sustained competitive advantage through technological progress and organisational efficiency. Using Penn World Tables data accessible via FRED (FRED, 2024), Figure 2 plots TFP evolution for Switzerland alongside the United States, Germany, and France, indexed to 2021 = 1.

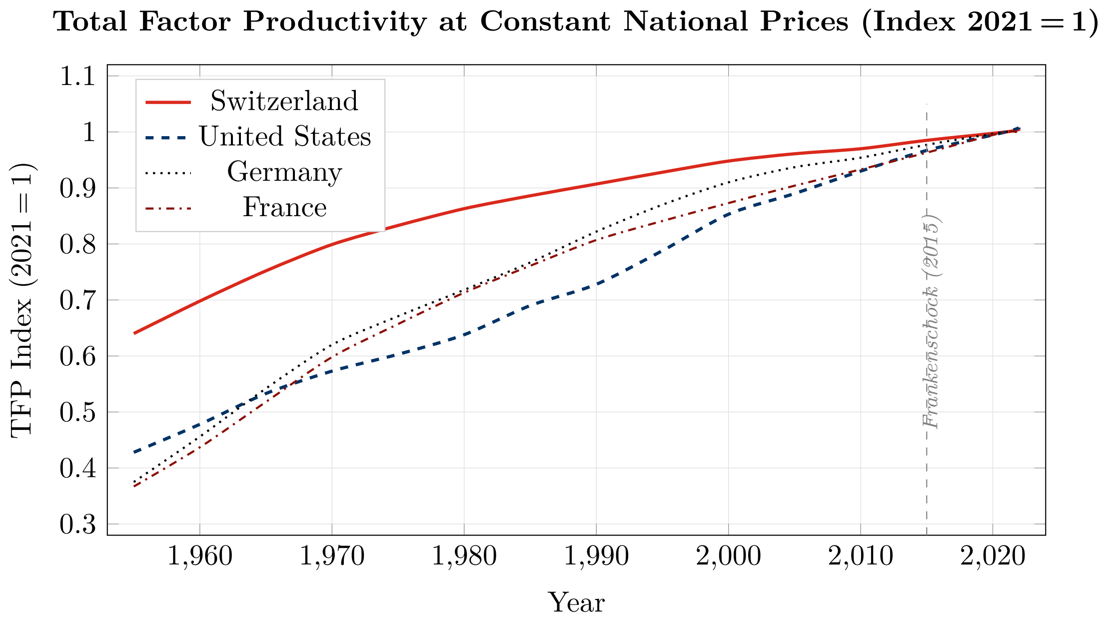

Figure 2 TFP evolution: Switzerland, United States, Germany, France (2021 = 1). Switzerland starts from a high absolute level but exhibits slower subsequent TFP growth, consistent with the "productivity frontier trap."
FRED (2024); Penn World Tables via FRED.

The trajectory reveals a critical structural feature of Swiss productivity dynamics. Switzerland began the post-war period with a markedly higher TFP *level* than its peers—a legacy of wartime industrial neutrality, early specialisation in precision chemical and mechanical industries, and deep human capital accumulation. However, subsequent decades display a relative slowdown: while the United States, Germany, and France experienced rapid TFP growth driven by catch-up imitation and capital deepening (the Solow growth engine), Switzerland—already on the global frontier—had limited scope for catch-up growth and consequently exhibited lower TFP growth *rates*.

This phenomenon—the "productivity frontier trap" (Acemoglu et al., 2006)— implies that frontier economies must innovate rather than imitate to sustain TFP growth. Switzerland's strategic response, heavily investing in R&D, human capital, and innovation infrastructure, reflects precisely this imperative. The 2015 *Frankenschock* (SNB exchange rate floor removal) created a measurable downward pressure on TFP as export-oriented firms absorbed currency losses, though the long-run adaptive response—automation, product upgrading, natural hedging—partially restored the growth trajectory.

2.2 Labour Productivity: International Benchmarking

While TFP captures the efficiency of all factors combined, labour productivity (output per hour worked) provides a more transparent measure of competitive strength. Figure 3 benchmarks Switzerland against key economies using 2024 data (FRED, 2024; OECD, 2024).

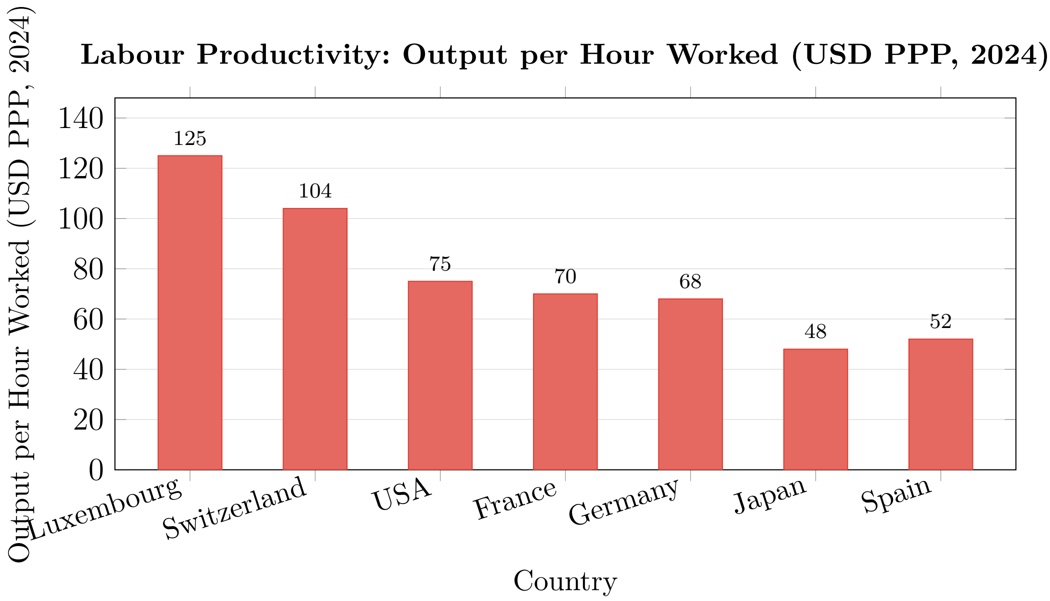

Figure 3 Labour productivity benchmarking (output per hour worked, USD PPP, 2024). Luxembourg is an outlier driven by hypertrophied financial sector weight. Switzerland ranks second, substantially outperforming the US and the largest EU economies.
FRED (2024); OECD (2024).

At \$104 per hour worked, Switzerland ranks second globally (excluding Luxembourg whose extreme outlier status reflects financial-sector distortion) (FRED, 2024). This figure substantially exceeds the United States (\$75), France (\$70), and Germany (\$68). High labour productivity is sustained by: *(i)* a highly skilled workforce—43% of the adult population holds a university degree (OFS, 2023); *(ii)* sustained capital deepening driven by domestic and foreign investment; *(iii)* sectoral specialisation in knowledge-intensive industries with high output-to-worker ratios; and *(iv)* the dual vocational education system that produces workers with immediately usable technical competencies.

2.3 R&D Investment: The Innovation Engine

Switzerland's commitment to research and development is exceptional by international standards. Figure 4 presents R&D expenditure as a share of GDP for selected OECD economies in 2021.

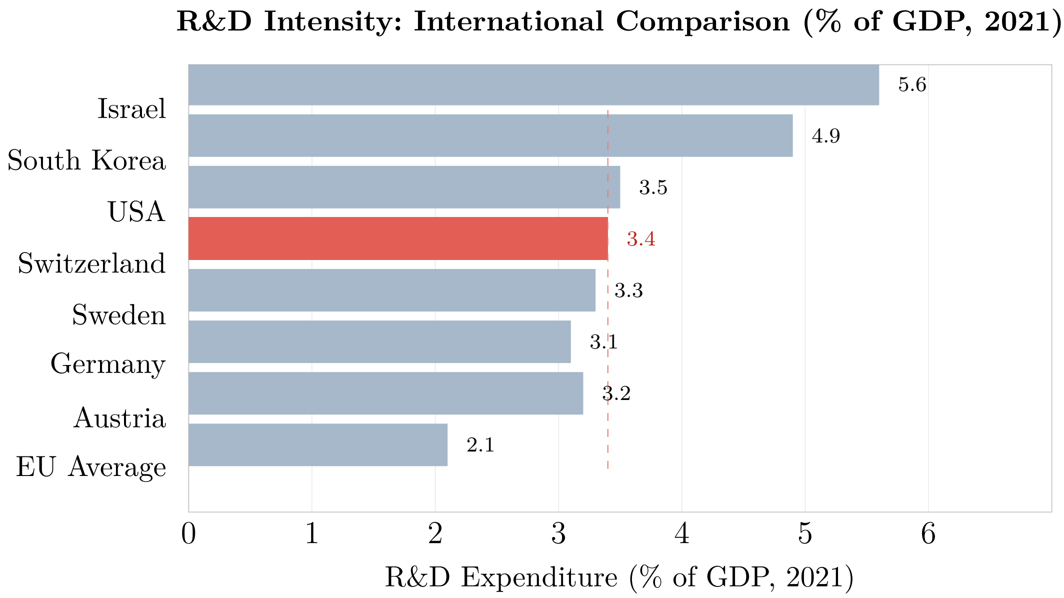

Figure 4 R&D intensity across selected OECD economies. Switzerland (red bar, 3.4% of GDP) ranks 4th, above the EU average of 2.1%. The dashed line marks Switzerland's level for direct visual comparison.
OFS (2023).

At 3.4% of GDP (CHF 24.6 billion in 2021), Switzerland's R&D intensity ranks fourth among OECD nations (OFS, 2023). Three features of this investment merit particular attention. *First*, the private sector finances 86% of its own R&D activities internally, demonstrating exceptional enterprise-level commitment without state dependency. The pharmaceutical sector alone (e.g.,Novartis and Roche) contributed CHF 6.247 billion in 2021—37% of total private enterprise R&D expenditure. *Second*, 42% of the 139,431 individuals employed in R&D are foreign nationals, reflecting both the attractiveness of Swiss research institutions (ETH Zurich, EPFL, University of Basel) and a permissive immigration framework via EU bilateral agreements for high-skilled workers (OFS, 2023). *Third*, higher education institutions contribute CHF 6.925 billion (28% of the national total), creating a virtuous public-private knowledge spillover dynamic: the non-rival character of knowledge (Romer, 1990) ensures that university research generates positive externalities across the private innovation ecosystem.

2.4 Patent Density: Translating Innovation into Appropriable Returns

Innovation investment translates into competitive advantage through intellectual property protection. Switzerland leads the world in patent applications per capita, filing 1,140 applications per million inhabitants—substantially exceeding key European peers like Germany (250), though raw comparisons with the United States and Japan at the EPO must account for their domestic filing preferences (EPO, 2024). In 2024, Swiss applications at the European Patent Office (EPO) reached 9,966, a 3.2% year-on-year increase.

Table 1 Swiss Patent Applications by Technology Field (EPO, 2024)
EPO (2024). Switzerland: 1,140 applications per million inhabitants (#1 globally).

Rank	Technology Field	Share (%)	YoY Growth
1	Medical Technology	10.5	+4.1%
2	Measurement & Precision Instr.	9.1	+2.8%
3	Consumer Goods (incl. watches)	8.3	+1.9%
4	Electrical Machinery & Energy	8.2	+8.9%
5	Pharmaceuticals	4.7	+3.6%
6	Organic Chemistry	4.6	+2.1%
7	Biotechnology	4.3	+5.2%
	Other fields	50.3	—
	Total applications (9,966)	**100%**	**+3.2%**

The sectoral distribution mirrors the export specialisation examined in Section 4: medical technology (10.5%), measurement and precision (9.1%), and the fast-growing electrical machinery sector (8.2%, +8.9%) dominate, confirming that Swiss patent activity is commercially oriented rather than academically self-referential.

2.5 Human Capital: The Dual Education Advantage

The Swiss vocational education and training (VET) system—whereby 58% of Swiss youth aged 15–19 pursue apprenticeships across 230 formally recognised professions—constitutes one of the most distinctive institutional features of the Swiss economy (OFS, 2023). This system achieves two objectives simultaneously: *(i)* it ensures a substantial portion of the workforce possesses highly specific and immediately applicable technical skills aligned with employer needs, minimising skill mismatch; and *(ii)* it reduces youth unemployment (7.6% in Switzerland versus an EU average of approximately 15%), generating social cohesion benefits that reinforce labour market flexibility.

This permeable human capital structure — where VET graduates frequently progress to Universities of Applied Sciences to complement the research scientists and managers produced by elite universities — supplies precisely the talent mix demanded by pharmaceutical manufacturers, precision engineering firms, financial institutions, and high-tech startups.

3. Industry-Level Determinants of Competitiveness

3.1 Revealed Comparative Advantage and Structural Specialisation

A fundamental insight from Balassa (1965) is that true comparative advantage is revealed not by theoretical factor endowments but by actual export performance. The Balassa Revealed

Comparative Advantage (RCA) index,

$$\mathrm{RCA}_{ij} = \frac{X_{ij}/X_i}{X_{wj}/X_w},$$

compares the share of product j in country i's total exports to the share of product j in world exports. An RCA > 1 denotes comparative advantage. For Switzerland, RCA values substantially exceed unity in pharmaceuticals, precious metal refining, precision instruments, watchmaking, and high-end chemicals, whereas the country exhibits low RCA in automobiles, heavy industry, and standardised consumer electronics—all capital-intensive or scale-sensitive sectors in which larger economies derive fixed-cost advantages.

Table 2 Top 5 Swiss Export Products: Value, Share, and Revealed Comparative Advantage (2023)
World Bank (2023); RCA estimates based on author calculations.

#	Product (HS6 Category)	Value (bn USD)	Share	RCA
1	**Gold, unwrought (non-monetary)** *Refining precision; Swiss refineries Valcambi, PAMP*	97.8	23.3%	$\gg$1
2	**Human/animal blood; microbial cultures** *Roche & Novartis biotech R&D ecosystem*	51.96	12.4%	$\gg$1
3	**Medicaments — mixed/unmixed products** *Basel pharma cluster; world-leading patent portfolio*	40.61	9.7%	$\gg$1
4	**Heterocyclic compounds (N-heteroatoms)** *Pharma & agrochemical active ingredient intermediates*	14.72	3.5%	>1
5	**Wrist-watches with automatic winding** *"Swiss Made" luxury brand; precision mechanics heritage*	14.39	3.4%	$\gg$1
	Total merchandise exports (all products)	*419.9*	*100%*	—

3.2 The Post-Industrial Processing Model in Detail

Switzerland's most distinctive structural feature is its *post-industrial processing model*: the systematic importation of raw or semi-processed inputs, their transformation through knowledge-intensive processes, and re-exportation at dramatically higher unit values. This model renders competitive advantage independent of natural resource endowments—a critical strategic property.

Gold refining exemplifies the model most clearly. Switzerland imports unrefined *doré* bars from mines across Africa, South America, the United States, and Australia; scrap gold from jewellers and e-waste processors; and large 12.5 kg bars from Western central banks. Four world-class Swiss refineries— Valcambi, PAMP, Argor-Heraeus, and Metalor—transform these inputs into 99.99% pure gold bars exported to China and India (kilobars for jewellery), the Middle East (kilobars and gold jewellery), and central banks globally for reserve management. In 2023, gold imports reached \$101.5 billion while refined gold exports reached \$97.8 billion, generating refining margins through value addition rather than commodity extraction (World Bank, 2023). The apparent gap between gold imports (101.5$bn)andexports($97.8bn) does not indicate a commercial loss; rather, it reflects Switzerland's dual role as both a global refining hub and a premier financial safe haven. A significant portion of imported gold is retained within the Swiss financial system as institutional reserves or private wealth assets, while the refining industry itself captures stable margins through high-purity processing and value addition.

Pharmaceuticals and biotechnology replicate this logic at higher technological complexity. Switzerland imports active pharmaceutical ingredients (APIs) and biological inputs—human/animal blood, microbial cultures: \$17.1 billion in imports in 2023—transforms them through sophisticated formulation, clinical validation, and quality-controlled GMP manufacturing, and re-exports finished pharmaceuticals (\$40.61 billion) and biological therapeutics (\$51.96 billion). The resulting value-added ratios far exceed those achievable in standardised manufacturing.

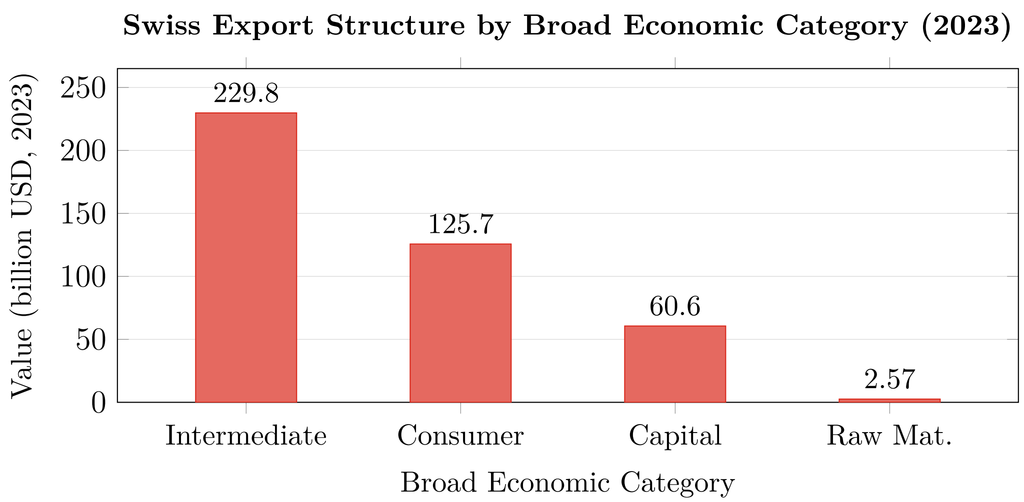

Figure 5 Swiss export structure by broad economic category, 2023. Intermediate goods dominate at 54.7% (\$229.8 bn), consistent with the post-industrial processing model.
World Bank (2023).

3.3 Sectoral Productivity Differentials and Structural Shifts

OECD STAN data permits broad cross-sectoral productivity comparisons for Switzerland. The chemical and pharmaceutical sector generates output per worker approximately 3–4 times the manufacturing average, while precision instruments and watchmaking operate at 2–3 times the average. This is consistent with the Melitz-type prediction that only the most productive firms enter export markets: Swiss pharmaceutical and instrument firms that survive international competition must be, by revealed preference, among the most efficient producers in their global industry.

Over the 1990–2023 period, Switzerland underwent a meaningful structural shift away from standardised textile and basic metalworking activities—sectors where Swiss wages were uncompetitive—towards higher-value pharmaceuticals, medtech, and financial services. Employment shares in pharmaceuticals and medtech have grown substantially, while textile employment has declined by over 70% since 1990. This structural upgrading, driven partly by franc appreciation pressures forcing a reallocation of capital and labor towards more productive sectors (Melitz, 2003), partly by deliberate cluster policy at the cantonal level, has raised economy-wide average productivity.

4. International Trade: Integration, Structure, and Agreements

4.1 Aggregate Trade Flows and Geographic Distribution

Switzerland's total exports in 2023 amounted to $419.9 billion, reaching 225 countries and territories—a global reach extraordinary relative to its population of 8.9 million (World Bank, 2023). Total imports were $365.4 billion, yielding a merchandise trade surplus. The trade-to-GDP ratio for goods and services combined substantially exceeds 100%, confirming Switzerland's classification as one of the world's most open economies.

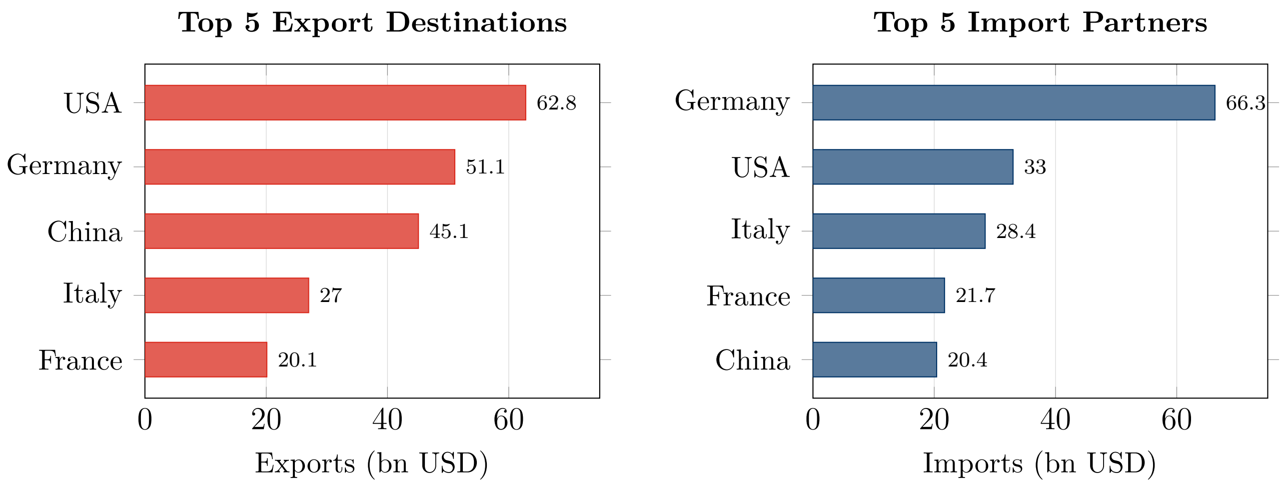

Figure 6 Top 5 Swiss export destinations (left) and import partners (right), 2023 (bn USD). The symmetric presence of Germany and the US on both sides reflects deep intra-industry trade in pharmaceutical and chemical intermediates.
World Bank (2023).

The geographic distribution reflects proximity and specialisation effects simultaneously. The United States, the primary export destination at $62.8 billion (14.97%), absorbs principally pharmaceuticals and precision instruments, reflecting deep US-Swiss pharmaceutical supply chain integration. Germany ($51.1 billion, 12.18%) is tightly integrated via intermediate goods supply chains—a natural consequence of Switzerland's geographical centrality within the Central European manufacturing hub. China ($45.1 billion, 10.76%) absorbs gold, luxury watches, and high-tech equipment, a composition substantially transformed by the 2013 Sino-Swiss FTA discussed in Section 6.

The symmetric presence of Germany and the United States as both top export destinations and top import sources implies substantial intra-industry trade, consistent with the Grubel-Lloyd hypothesis of vertical and horizontal product differentiation (Grubel and Lloyd, 1975): Switzerland exports finished pharmaceuticals to the US while importing APIs; it exports precision instruments to Germany while importing machinery and automobiles.

4.2 Tariff Policy and WTO Integration

Switzerland has been a GATT/WTO member since 1995, and its applied Most-Favoured-Nation (MFN) tariff rates reflect a highly liberalised trade regime for industrial goods, though with notable agricultural peaks reflecting domestic food policy.

Table 3 Switzerland: Applied MFN Tariff Rates by Broad Sector (2022)
WTO (2023). Agricultural tariff peaks applied as specific and compound duties.

Sector	**Simple Avg. MFN (%)**	**Trade-wtd Avg. (%)**	**Key Note**
Agriculture (live animals, food, cereals)	28.4	15.2	Peak tariffs on sugar/cereals
Textiles and clothing	4.2	2.8	Declining under bilateral deals
Chemicals and pharmaceuticals	0.0	0.0	Swiss zero-tariff pharma policy
Machinery and mechanical equipment	0.0	0.0	Capital-good liberalisation
Precision instruments and watches	0.5	0.3	Near-zero
Precious metals and jewellery	0.0	0.0	Gold-refining strategy
All merchandise (weighted avg.)	**8.6**	**2.1**	Agriculture inflates simple avg.

The near-zero tariffs on pharmaceuticals, machinery, and precious metals reflect deliberate Swiss industrial strategy: by eliminating import duties on production inputs, Switzerland reduces the marginal cost of its refine-and-re-export model, effectively subsidising the value-addition chain. Agricultural tariff peaks— reflecting domestic political economy rather than competitiveness strategy— represent the major distortion and have been subject to persistent OECD scrutiny as sources of consumer welfare loss.

4.3 Regional Trade Agreements: EFTA, EU Bilaterals, and Beyond

Switzerland's trade governance architecture is among the most complex in the world, layering WTO multilateral commitments with multiple preferential regimes.

EFTA. Switzerland is a founding EFTA member (1960). EFTA has concluded FTAs with 30 FTA partners covering 41 countries and territories outside the EU, including the Gulf Cooperation Council, Central American states, and several Asian economies, providing Swiss goods preferential access in markets where EU agreements do not apply.

Bilateral agreements with the EU. Following the 1992 EEA referendum rejection, Switzerland negotiated over 120 bilateral sectoral agreements with the EU, covering the free movement of persons, air transport, land transport, public procurement, mutual recognition of standards, research cooperation (Horizon Europe), and agriculture. These agreements provide Swiss exporters with near-Single-Market access without formal EU membership. The EU remains Switzerland's dominant trade partner, absorbing approximately 48% of exports and supplying approximately 67% of imports.

Independent Swiss FTAs. Beyond EFTA's network, Switzerland has independently negotiated FTAs with Japan (2009), and—most significantly—China (2013). These agreements complement EFTA's reach, giving Swiss firms preferential access to strategic non-EU markets that would otherwise be subject to standard MFN rates.

5. Structural Policies and Their Empirical Impact on Competitiveness

5.1 SNB Monetary Policy: The *Frankenschock* as Natural Experiment

The Swiss franc's safe-haven status—persistently elevated during periods of global risk aversion—creates structural competitiveness challenges for Swiss exporters priced in foreign currencies. The *Frankenschock* has occurred twice in the modern era.

The *first shock* materialised in August–September 2011 when EUR/CHF briefly approached parity. The SNB responded on 6 September 2011 by imposing an exchange rate floor of CHF 1.20 per euro—promising unlimited foreign asset purchases to maintain the cap. This ceiling protected Swiss exporters for 40 months but at the cost of a rapidly expanding SNB balance sheet.

The *second shock*, on 15 January 2015, arose when the SNB unexpectedly removed the exchange rate floor, triggering immediate CHF appreciation of 15–20% relative to the euro. Simultaneously, the SNB cut the policy rate to -0.75%—the lowest in the world at that time—to partly offset the competitiveness blow.

Eufinger et al. (2021) exploit this as a natural experiment, using a *difference-in-differences* (DiD) identification strategy: Swiss firms constitute the treatment group (affected by the rate cut and CHF shock), while comparable German firms—facing no equivalent monetary policy shift—constitute the control group. The DiD estimator identifies the causal effect of the SNB policy on corporate outcomes.

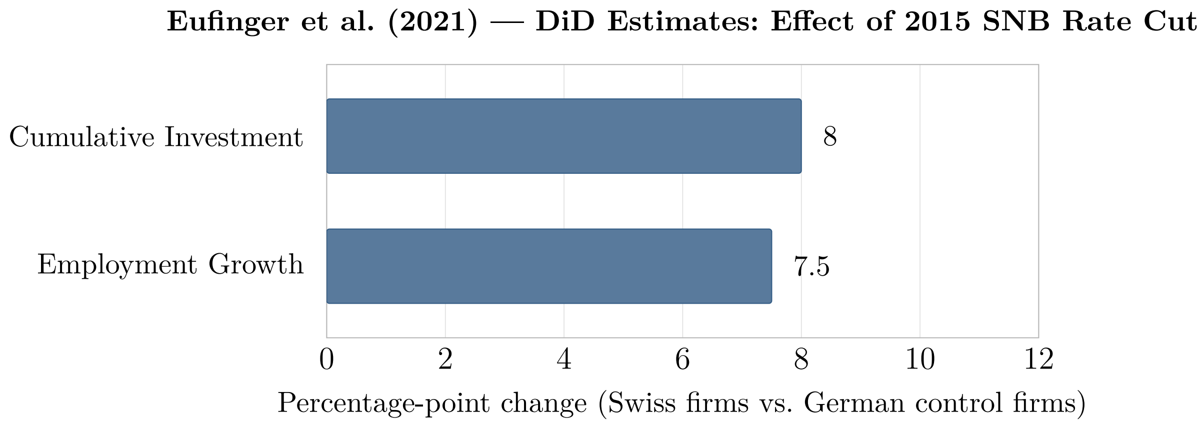

Figure 7 Difference-in-differences estimates: 2015 SNB rate cut (to −0.75%) on Swiss firm outcomes relative to German control firms. Cumulative investment +8 pp; employment +7.5 pp.
Eufinger et al. (2021).

The results of Eufinger et al. (2021) show that following the rate cut:

- Cumulative firm investment increased by approximately **8 percentage points** relative to control firms.
- Employment growth increased by approximately **7.5 percentage points** relative to control firms.
- Effects were concentrated among financially constrained firms with limited internal cash flows, consistent with standard financial accelerator theory.

While the study does not directly estimate the effect on export volumes, the investment-productivity link established in Melitz (2003) implies that investment-driven efficiency gains translate into expanded exporter market share over time, as firms improved their efficiency to stay competitive despite the high value of the currency. The long-run adaptive response of Swiss exporters to franc strength— automation and capital deepening, product-mix upgrading towards even higher-margin patented goods, and natural hedging through geographic cost-revenue matching— reinforced, rather than weakened, the knowledge intensity of Swiss industry.

5.2 Innovation Policy: Public R&D Subsidies and the Innosuisse Framework

The causal impact of public innovation support on firm-level performance is studied rigorously by Arvanitis et al. (2010), whose analysis centres on subsidies distributed by Switzerland's Commission for Technology and Innovation (CTI, subsequently rebranded Innosuisse).

Identification challenge. Subsidised firms may be systematically more innovative ex ante, creating upward selection bias if treated and untreated firms are simply compared in levels. Arvanitis et al. address this through *Propensity Score Matching (PSM)*: they estimate the probability of receiving a subsidy as a function of observable pre-treatment characteristics (firm size, industry, prior R&D intensity, innovation activity), then construct a matched control group of observationally similar non-subsidised firms, While this method helps reduce selection bias, it mainly accounts for visible factors like firm size, meaning some hidden differences between firms may still exist.

Results. After controlling for selection into treatment, subsidised firms exhibit: *(i)* significantly higher R&D intensity (innovative inputs); *(ii)* stronger innovation outputs (new products, process innovations); and *(iii)* a higher share of sales attributable to innovative products.

The established causal chain is:

$$\text{Public R\&D Subsidies} \rightarrow \text{Higher Firm R\&D} \rightarrow \text{More Innovation Outputs}$$
$$\rightarrow \text{Stronger Firm Competitiveness.}$$

This finding validates the Swiss government's innovation promotion strategy and provides empirical justification for continued Innosuisse funding—particularly as fiscal pressures from the OECD Pillar Two minimum tax may force a rebalancing from tax-based (IP Box, participation exemption) towards subsidy-based (direct grants, matching funds) industrial policy instruments.

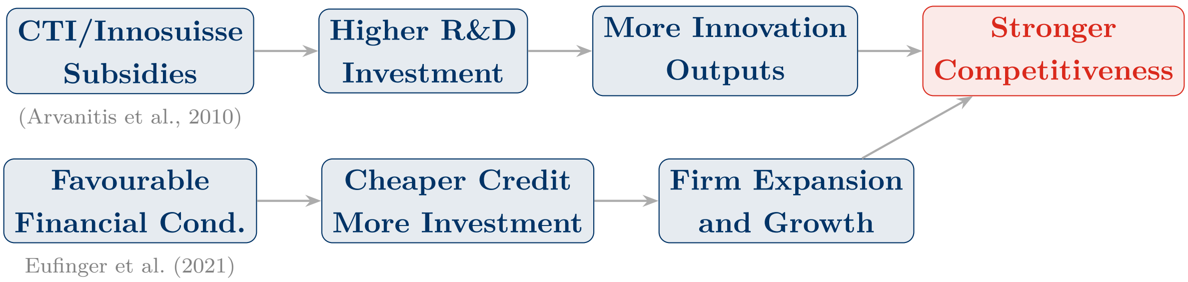

Figure 8 Structural policy transmission channels to firm competitiveness. Top row: innovation policy channel (Arvanitis et al., 2010). Bottom row: financial conditions channel (Eufinger et al., 2021). Both converge at the same endpoint.

5.3 Labour Market Flexibility and Institutional Complementarities

Switzerland's labour market is characterised by high flexibility relative to continental European comparators. Collective bargaining occurs predominantly at the industry (rather than firm or national) level, allowing wage setting to reflect sector-specific productivity while retaining economy-wide adjustment capacity. While several cantons introduced cantonal minimum wages post-2020, there is no statutory national minimum wage, and hiring and firing regulations are relatively light compared to France or Germany.

This flexibility has contributed to Switzerland's capacity to absorb macroeconomic shocks—including both *Frankenschock* episodes and the COVID-19 pandemic—without generating persistent unemployment. The *Kurzarbeit* (short-time work) scheme allowed firms to retain trained workers during temporary demand contractions, preserving human capital and avoiding the long-term damage that plague less flexible labour markets. The low overall unemployment rate (2.9%, ILO measure) and low youth unemployment (7.6%) together testify to the institutional efficiency of this system (OFS, 2023).

6. The Role of China: Import Competition, Market Access, and the FTA

6.1 China's WTO Accession and the Import Competition Channel

China's WTO accession in December 2001 fundamentally altered the global trade landscape, exposing domestic producers in advanced economies to intensified low-cost competition—a dynamic extensively documented by Autor et al. (2013) (the "China shock") in the context of United States manufacturing employment. The China shock operated through two principal channels: *direct* import competition displacing domestic production in home markets, and *indirect* competition in third-country export markets.

For Switzerland, the direct import competition channel was structurally limited. Swiss exports are almost entirely concentrated in highly differentiated, knowledge-intensive, and patent-protected goods—pharmaceuticals, precision instruments, luxury watches—where price-based competition from China is less effective, especially in high-end segments, although it created significant pressure on lower-tech Swiss manufacturing in earlier decades. Swiss pharmaceutical firms do not compete with Chinese API manufacturers on cost; they compete on innovation, patents, regulatory approval history, and clinical efficacy profiles. Chinese imports into Switzerland reached $20.4 billion in 2023 (5.58% of total imports), concentrated in electronics, textiles, and parts—categories largely *complementary* to Swiss production rather than competitive substitutes (World Bank, 2023). The absence of a significant "Swiss China shock" in manufacturing employment confirms this insulation.

6.2 The 2013 Sino-Swiss Free Trade Agreement: Architecture and Effects

Signed on 6 July 2013 and entering into force on 1 July 2014, the Sino-Swiss FTA was the first free trade agreement concluded between China and a continental European country—a significant diplomatic and economic achievement reflecting Switzerland's commercial neutrality and strategic positioning.

The agreement's economic provisions were asymmetric in structure:

- **Chinese commitments:** China agreed to eliminate tariffs on 99.7% of Swiss export value over transition periods of 5, 10, or 15 years. Key beneficiaries include pharmaceuticals (tariffs of 3–6% eliminated), precision instruments, watches, and chemical products.
- **Swiss commitments:** Switzerland agreed to eliminate tariffs on 99.9% of Chinese export value. Given Switzerland's already near-zero MFN tariffs on manufactures, this commitment was largely confirmatory of existing practice.
- **Investment and IPR provisions:** The agreement incorporated investment protection, strengthened intellectual property rights provisions (aligned with Swiss patent interests), and trade facilitation measures addressing customs procedures and border clearance efficiency.

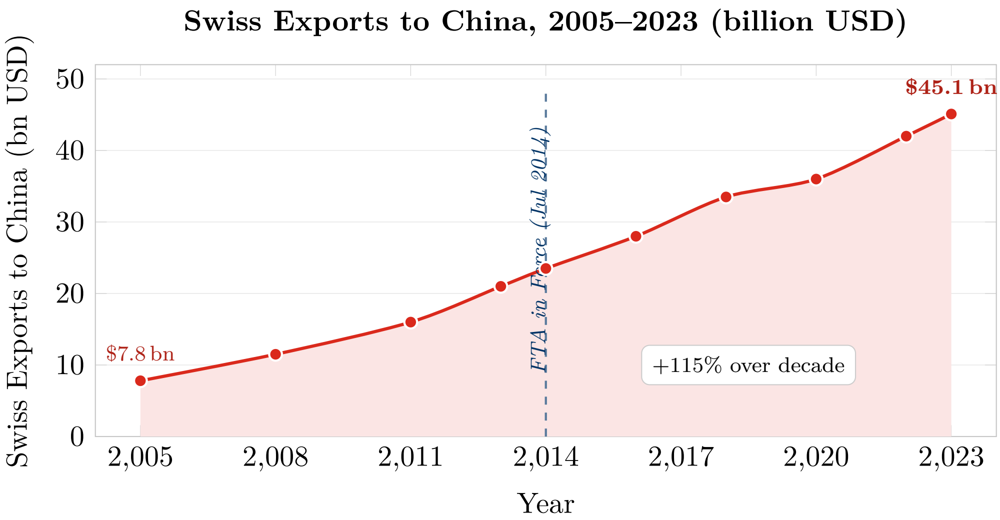

Figure 9 Swiss exports to China, 2005–2023 (bn USD). Area chart shows the sustained growth trajectory; the FTA (July 2014) formalised and accelerated the upward trend, with exports nearly doubling in the following decade.

World Bank (2023); pre-2020 estimates based on Swiss Federal Customs data.

The *market access channel* has been economically substantial. Swiss exports to China grew from approximately \$21 billion in 2013 to \$45.1 billion in 2023—a 115% increase over a decade, substantially outpacing Switzerland's overall export growth (World Bank, 2023). The composition—gold, luxury watches, precision instruments, and pharmaceuticals—reflects the FTA's success in formalising and deepening trade relationships in categories where Switzerland holds strong and durable comparative advantage.

6.3 Shifting Supply Chains, Geopolitical Risks, and Counterfeiting

While the FTA has expanded Swiss market access in China, it has also deepened supply chain dependencies that carry strategic risk. Switzerland imports \$20.4 billion of goods from China annually, with electronics, textiles, and components constituting the bulk. Geopolitical tensions between China and Western allies—combined with supply chain disruptions revealed during the COVID-19 pandemic—have prompted Swiss policymakers to assess vulnerabilities in critical supply chains, particularly pharmaceutical APIs (where China and India jointly dominate global production), electronic components, and rare earth materials essential for precision manufacturing.

The FTA's intellectual property provisions face persistent enforcement challenges in practice. Swiss watchmakers continue to experience grey-market and counterfeit competition originating in China, with counterfeit Swiss watches constituting a meaningful fraction of seizures at European customs points (WTO, 2023). The long-term reputational risk to the "Swiss Made" label—a key component of the luxury watch premium—requires ongoing bilateral engagement on IPR enforcement.

6.4 The "China Effect" on Swiss Competitive Positioning: Net Assessment

On balance, China's rise has been a *net positive* for Swiss competitiveness, operating primarily through the market access channel rather than the import competition channel. The structural complementarity of Chinese demand—for luxury goods, refined gold, pharmaceutical products, and precision instruments—with Swiss export specialisation has generated substantial and durable export growth. The Sino-Swiss FTA formalised and accelerated this dynamic, while the EFTA framework provided a multilateral backdrop ensuring most-favoured treatment for non-FTA EFTA members.

The medium-term competitive challenge from Chinese high-technology upgrading— particularly in biopharmaceuticals (emerging Chinese CDMO firms), automation and robotics (threatening Swiss precision machinery niches), and electric vehicle supply chains (affecting Swiss precision component exporters)—constitutes a genuine but manageable risk, provided Switzerland continues to invest at the innovation frontier. The Arvanitis et al. (2010) evidence suggests that public innovation subsidies can causally support this continued frontier investment even as traditional fiscal instruments are constrained by the OECD Pillar Two framework.

7. Conclusion

Switzerland's economic competitiveness represents one of the most instructive case studies in modern development economics: a resource-poor, politically neutral small open economy that has sustained frontier-level productivity, living standards, and export performance through decades of consistent investment in knowledge intensity, institutional quality, and adaptive structural policy design.

This report has established four main conclusions. **First**, Switzerland's TFP trajectory reflects the dual challenge of the productivity frontier: having reached a high absolute level early in the post-war period, Switzerland faces structurally lower TFP *growth rates* than catching-up economies. Its strategic response—R&D investment at 3.4% of GDP, world-leading patent density, and a human capital system producing both technical and scientific talent—constitutes the appropriate frontier-economy competitive strategy.

Second, the post-industrial processing model renders Switzerland's competitive advantage largely immune to traditional factor cost competition. By specialising in knowledge-intensive transformation—gold refining to 99.99% purity, pharmaceutical formulation at GMP standards, biotech manufacturing of patented biologics, luxury watchmaking with brand heritage—Switzerland competes in market segments where quality, precision, regulatory compliance, and brand reputation determine success, not labour cost arbitrage. This structural insulation explains why Chinese industrialisation has been a net market access opportunity rather than a competitiveness threat.

Third, Swiss structural policies have empirically demonstrable and causally identified effects on firm competitiveness. Arvanitis et al. (2010) show, through propensity score matching on the Swiss Innovation Survey, that CTI subsidies causally raise firm R&D intensity, innovation output, and the share of revenues derived from innovative products. The SECO natural experiment (Eufinger et al., 2021), exploiting the 2015 SNB policy shock via difference-in-differences, demonstrates that favourable financial conditions raise cumulative investment by 8 percentage points and employment by 7.5 percentage points relative to control firms. Both results converge on the conclusion: well-designed structural policies—innovation subsidies and financial system openness—materially and causally enhance Swiss firm competitiveness.

Fourth, the 2013 Sino-Swiss FTA has been economically successful, supporting a 115% growth in Swiss exports to China over a decade. The medium-term challenge from Chinese high-technology upgrading in biopharmaceuticals, automation, and advanced manufacturing requires sustained attention and continued R&D investment to maintain Switzerland's innovation lead.

The "Swiss Formula"—*Smart Policy → R&D and Capital → Innovation → Productivity → Export Competitiveness*—constitutes a replicable structural template for small open economies navigating the knowledge economy. Its long-run sustainability depends on maintaining institutional quality, educational excellence, and the political will to continue investing in the innovation frontier even as the OECD Pillar Two minimum tax regime gradually constrains traditional fiscal policy instruments.

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